

Hyun-Jung Kim

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Professional Profile

Computational materials physicist with 10+ years of experience in **electronic-structure theory, quantum-chemical simulation, Hamiltonian construction / numerical eigensolver workflows**, and **electronic materials / OLED R&D**, followed more recently by enterprise AI enablement and workstream leadership. At LG Display, I connect research depth with operational execution through **AI project review, tech scouting, internal AI capability building, GitLab-based MLOps platform operation, LLM / agentic workflow automation**, and lightweight web services for education and event participation in a security-constrained industrial environment. I support repository-based management of AI projects, algorithms, and deliverables, helping centralize reusable outputs and establish governance-aligned work routines for manufacturing AI/R&D teams.

Selected Highlights

- **Scientific R&D depth:** 10+ years in computational materials physics, DFT, quantum chemistry, multiscale electronic-structure modeling, and Hamiltonian/eigensolver research software; **33 peer-reviewed journal articles and one peer-reviewed paper in the SID Symposium Digest of Technical Papers**, including **Physical Review Letters, Physical Review B, Physical Review Materials, ACS journals**, and **npj Computational Materials**.
- **Electronic materials / OLED:** quantum-chemical and DFT-driven analysis for OLED-relevant molecular/materials design, structure-property understanding, and ML-assisted inverse-design framing, including **blue TADF emitter** strategy work presented and published at **SID Display Week 2024**.
- **Enterprise AI / GenAI enablement:** current workstream lead in the **AI Governance Team** at LG Display, covering **AI project review, portfolio management, tech scouting, GitLab-MLOps platform operation**, and secure internal AI adoption in an on-premise environment.
- **Quantum-technology coordination:** serve as **LG Display's coordinator** in an **LG Group quantum-technology working group**, reviewing quantum-computing technologies, defining candidate application problems, and helping advance selected topics toward project formulation.
- **Repository-centered AI transformation support:** supported centralized management of AI algorithms, project repositories, technical deliverables, and collaboration workflows through GitLab-based MLOps operations, enabling more traceable project execution and faster adoption of agentic workflow practices.

Education

- **Ph.D. in Theoretical Condensed Matter Physics, Hanyang University**, Mar. 2011 – Aug. 2015
Advisor: Prof. Dr. Jun-Hyung Cho
Thesis: *Electronic properties of self-assembled low-dimensional nano-structures on surfaces*
- **M.S. in Theoretical Condensed Matter Physics, Hanyang University**, Mar. 2009 – Feb. 2011
Advisor: Prof. Dr. Jun-Hyung Cho
Thesis: *Length- and parity-dependent electronic states in one-dimensional carbon chains on C(111)*
- **B.S. in Physics, Hanyang University**, Feb. 2003 – Feb. 2009
Thesis: *First-principles study of the structure and electronic properties of C(110) surface*; military service, Jan. 2004 – Jan. 2006

Research & Industrial Experience

Industry positions

- **Sep. 2022 – present: Senior Research Engineer / Workstream Lead**,
AI/Big Data Research Division, AX Group, LG Display, Paju Complex 245, LG-ro, Wollong-myeon, Paju-si, 10845 Gyeonggi-do, Korea
Role:
 - **OLED Materials R&D (2022–2024)**: Performed quantum-chemical and DFT-based analysis of OLED-relevant molecules and materials, translating excited-state and structure–property insights into **molecular design**, **screening**, and **inverse-design** problem framing; contributed to ML strategy work for **blue TADF emitters** presented and published at **SID Display Week 2024**.
 - **AI Instructor (2025)**: Designed and delivered internal training on **AI/ML fundamentals**, **workflow automation**, and **LLM application development**; built and shared web-based training materials for **executive AX education** and **Git/GitLab onboarding**; contributed to **AX certification** education and **executive AI capability-building** sessions.
 - **AI enablement and workflow tool builder (recent)**: Built and operated lightweight web platforms for internal education and company-wide participation, including a **Cloudflare Workers / Durable Objects** real-time voting and audience-engagement system as one practical serverless hosting and repository-driven delivery example.
 - **GitLab-MLOps Platform Workstream (current)**: Operate and support a GitLab-based MLOps platform for an AI/Big Data / AX organization, covering repository-based AI project management, algorithm and deliverable tracking, collaboration workflow support, user onboarding, and governance-aligned usage in manufacturing AI/R&D teams.
 - **Workstream Lead (AI Governance Team; current)**: leading **AI project review and portfolio management**, **tech scouting**, and **on-premise / security-constrained AI adoption support**; helping centralize AI project outputs and guiding **GenAI / agentic workflow** use-case evaluation and internal enablement.
 - **LG Group quantum-technology working group (current)**: Serve as LG Display’s coordinator, reviewing quantum-computing technologies, defining candidate application problems, and helping advance selected topics toward project formulation.

Academic positions

- **Aug. 2020 – Jul. 2022: Visiting Scientist (supported by Humboldt Research Fellowship)**,
Peter Grünberg Institut (PGI-1) and Institute for Advanced Simulation (IAS-1), Forschungszentrum Jülich GmbH, D-52425 Jülich, Germany
Host: Prof. Stefan Blügel

- **Mar. 2020 – Jul. 2020: Postdoctoral researcher,**
Peter Grünberg Institut (PGI-1) and Institute for Advanced Simulation (IAS-1), Forschungszentrum Jülich GmbH, D-52425 Jülich, Germany
Advisor: Prof. Stefan Blügel
- **Sep. 2019 – Feb. 2020: Research fellow,**
Quantum Universe Center, Korea Institute for Advanced Study, Seoul, Korea
Advisor: Prof. Young-Woo Son
- **Sep. 2015 – Aug. 2019: Postdoctoral researcher,**
School of Computational Sciences, Korea Institute for Advanced Study, Seoul, Korea
Advisor: Prof. Young-Woo Son

Mentoring & Academic Service

- **M.Sc. thesis co-supervision (RWTH Aachen University / Forschungszentrum Jülich),**
Xiaoyu Liu, *“Electronic Structure of Twisted Bilayer Tungsten Diselenide investigated by a transferable Tight-Binding-Model”*.
Supervisor: Prof. Dr. Stefan Blügel
- **Ph.D. thesis committee member / examiner (POSTECH),**
Seung-Gyo Jeong, *“Topological defects in quasi one dimensional charge density wave”*.
Supervisor: Prof. Tae-Whan Kim

Short-Term Research Visits

- **Aug. 8 2022 ~ Aug. 31 2022,**
School of Computational Sciences, Korea Institute for Advanced Study (KIAS), Seoul, Korea,
Host: Prof. Young-Woo Son
- **Jun. 1 2018 ~ Nov. 30 2018,**
Peter Grünberg Institut (PGI-1) and Institute for Advanced Simulation (IAS-1), Forschungszentrum Jülich, Jülich, Germany,
Host: Prof. Stefan Blügel and Dr. Gustav Bihlmayer
- **Jul. 13 2015 ~ Aug. 31 2015,**
School of Computational Sciences, Korea Institute for Advanced Study (KIAS), Seoul, Korea,
Host: Prof. Young-Woo Son
- **Feb. 26 ~ Mar. 12 2015,**
ICQM, Peking University, Beijing, China,
Host: Prof. Ji Feng
- **Jan. 15 2015 ~ Jun. 30 2015: Visiting student,**
ICQD/HFNL, University of Science and Technology of China, Hefei, China,
Host: Prof. Zhenyu Zhang
- **Jan. 15 2013 ~ Jan. 27 2013,**
ICQD/HFNL, University of Science and Technology of China, Hefei, China,
Host: Prof. Changgan Zeng
- **Jul. 1 2011 ~ Sep. 30 2011: Visiting student,**
Peter Grünberg Institut and Institute for Advanced Simulation, Forschungszentrum Jülich, Jülich, Germany,
Host: Prof. Stefan Blügel
- **Dec. 1 2010 ~ Feb. 14 2011: Visiting student,**
Theory Department, Fritz-Haber-Institut der Max-Planck-Gesellschaft, Berlin, Germany,
Host: Prof. Matthias Scheffler

Honors, Awards & Fellowships

- *Humboldt Research Fellowship for Postdoctoral Researchers*
Alexander von Humboldt-Stiftung, 2020-2022
- *KIAS Academic Research Award*
Korea Institute for Advanced Study, September 20 2018
- *TJ Park (Cheong-Am) Science Fellowship*
POSCO TJ Park Foundation, 2018-2019
- *Outstanding PhD Dissertation Award*
Hanyang University, August 21 2015
- *Outstanding Paper Award*
Research Institute for Natural Sciences, Hanyang University, February 2015
- *Selected by the JSPS to participate in the "6th HOPE Meeting with Nobel Laureates"*
Japan Society for the Promotion of Science (JSPS), March 11-15 2014 (Tokyo, Japan)
- *Best Presentation Award (Oral)*
KPS Spring Meeting, The Korean Physical Society, April 2013
- *Hanyang Science and Technology Scholarship*
Hanyang University, Spring 2011 - Fall 2012
- *DAAD-NRF Summer Institute Program (Scholarship & travel grant)*
German Academic Exchange Service (Germany) and National Research Foundation (Korea), July-September 2011
- *Academic Excellence Scholarship*
Hanyang University, Spring 2007

Selected Research Funding

- *FZJ-NST Bilateral Cooperation Programme*
Tight-binding Hamiltonian parameterizations using artificial intelligence
Forschungszentrum Jülich (Germany) & National Research Council of Science and Technology (Korea), 2022
- *Humboldt Research Fellowship*
The role of strain and interlayer interaction in the layered higher-order topological insulators: Multi-scale numerical simulations
Alexander von Humboldt-Stiftung, 2020-2022 (Fellowship: 64,080 EUR)
- *Cheong-Am Science Fellowship*
A study on the novel properties of low-dimensional crystalline matter
POSCO TJ Park Foundation, 2018-2019 (Research fund: KRW 70 million)

Computational Skills

- **First-principles / quantum-chemistry codes:** FLEUR, VASP, FHI-aims, WIEN2k, AMS, Gaussian 16, ORCA
- **Programming / scientific computing:** Fortran, C, Python, MATLAB/Octave, Shell (bash); Hamiltonian construction, dense/sparse eigensolver workflows, parameter optimization, and C/Fortran/Python interface work in research software contexts
- **AI/ML & LLM stack (Python):** scikit-learn; PyTorch; gym (RL); LangChain / LangGraph
Focus: AI/ML-for-science prototyping, PyTorch-based RL experimentation for tight-binding parameter workflows, and RAG/MCP-oriented LLM application / agentic workflow prototyping

- **Developer / MLOps workflow:** Git; GitLab-MLOps platform operation; repository-based AI project and artifact management; GitLab-based collaboration and onboarding; CI/CD-oriented workflow enablement; Cloudflare Workers / Durable Objects deployment for real-time internal event services; governance-aligned usage support in security-constrained manufacturing AI/R&D environments
- **Selected AI workflow / training and control-plane assets:**
 - *GitLab-Onboarding-Lectures*: Git/GitLab onboarding curriculum for repository-centered collaboration, MR review, CI/CD quality gates, and self-managed GitLab operations.
 - *Lets_AX_EXE*: executive-facing GenAI / AX education and hands-on AI capability-building material.
 - *vibe-arena / Vibe Arena* and *agentic_vid / vid-gen*: supporting examples of Cloudflare serverless hosting and GenAI workflow harnessing, covering event views (wall, vote, message) and a local AI-agentic video workflow harness focused on Remotion scenes, Runway API wrappers, generation logs, review gates, budgets, and Codex-oriented orchestration.
 - *AI_Tech_Review* and *Federlicht*: evidence-based AI technology review, report generation, and research automation workspaces.
 - *CodexSkillszCline*: rule, workflow, and skill-pack assets for safer on-prem agentic workflow adoption.
 - *Task Memory Hub* (TMH; private / limited-access GitHub repository): local-first task and action memory hub with CLI, loopback REST API, STDIO MCP, global hub sync, agent registry, and harness-governed runner paths for safer agentic workflow operations.
- **Operating systems:** macOS, Linux, Windows
- **Open-source code development (selected):**
 - *TBFIT* (Fortran / Python): Slater–Koster tight-binding parameter fitting toolkit for constructing tight-binding Hamiltonians from geometry / parameter inputs and fitting them to first-principles band structures through Levenberg–Marquardt, Genetic Algorithm, and Particle Swarm Optimization workflows; includes dense/sparse eigensolver-oriented implementation paths and Python/PyTorch experimentation around band-data and RL-style parameter workflows (GitHub repository).
 - *VASPBERRY* (Fortran): Berry curvature / Chern number post-processing for VASP WAVECAR (GitHub repository).
 - *VASPBAUM* (Fortran): Band-unfolding machinery for VASP WAVECAR (k-projection), with optional circular dichroism; maintained as a reusable post-processing pipeline (GitHub repository).
 - *GitHub Arctic Code Vault Contributor* (2020).

Professional Education & Certificates

- **Post Graduate Program in Artificial Intelligence & Machine Learning,**
Caltech Center for Technology and Management Education (CTME), May 21, 2025, Simplilearn.
- **Quantum Programming Core,**
Certificate of Course Completion, August 15, 2025, D-Wave.

Research Overview

Research Interests (keywords): Density functional theory and beyond (vdW interactions, hybrid functionals, RPA, GW); electronic structure and quantum phenomena in low-dimensional systems; Hamiltonian

model construction and parameterization; spin-orbit coupling and topological materials; quantum magnetism; AI/ML for materials discovery; quantum computing for optimization and molecular design.

My research focuses on computational materials science, with a core emphasis on **molecular/materials design using first-principles and quantum-chemical simulations**, and on **AI/ML workflows** that translate simulation outputs into data-driven screening and inverse-design strategies. I bring a strong background in condensed-matter theory and electronic-structure methods, developed through studies of graphene, 2D transition-metal dichalcogenides, topological materials, and surface/adsorption systems, where I produced mechanistic insights into structure-property relationships, magnetism, and electronic/topological properties. I have published **33 peer-reviewed journal articles and one peer-reviewed paper in the SID Symposium Digest of Technical Papers**, including articles in **Physical Review Letters / Physical Review B / Physical Review Materials**, ACS journals, and **Nature Portfolio journals** (e.g., *Scientific Reports*; *npj Computational Materials*), and have presented at major international conferences.

During my postdoctoral work, I incorporated **global optimization** into model development, including robust parameterization of **tight-binding Hamiltonians** for electronic/topological properties to enable multi-scale simulations. This work involved scientific-software implementation around Hamiltonian construction, dense/sparse eigensolver workflows, and parameter fitting, shaping my approach to reliable and transferable computational workflows that couple physics-based models with modern optimization and learning techniques while maintaining interpretability and reproducibility.

In recent years, I have expanded my work toward **OLED-relevant molecular/materials design and mechanism analysis** in an industrial R&D environment. At LG Display, I apply quantum-chemical/DFT simulations and ML-enabled workflows to connect **structure-property relationships** to **emission and stability**, and to support data-driven material screening and design decisions. This includes contributing to ML strategies for **inverse design of blue TADF emitters**, trained on excited-state properties derived from DFT calculations and published in the *SID Symposium Digest of Technical Papers* after presentation at SID Display Week 2024.

In parallel, I am pursuing a forward-looking agenda on **hybrid quantum-classical approaches** for materials discovery. My current focus is on **problem framing and feasibility analysis**—for example, exploring how OLED-relevant quantum-chemical objectives could be reformulated into optimization forms compatible with quantum optimization paradigms (e.g., QUBO-style formulations), and how workflow prototypes could later be evaluated in scoped pilot projects. This direction is an ongoing effort and should be interpreted as preparation for future execution, rather than completed QC outcomes.

Going forward, my goal is to build an **OLED-targeted materials discovery pipeline** that unifies (i) state-of-the-art quantum chemistry for property prediction, (ii) AI/ML workflows for data-efficient modeling and active learning, and (iii) inverse design supported by **generative AI** and, where appropriate, **quantum optimization**, aligned with industrial strategies that aim to accelerate discovery-to-deployment for next-generation OLED materials.

Peer-Reviewed Publications: 33 Journal Articles + 1 SID Proceeding

American Physical Society Journals (23)

• Physical Review Letters

1. Sun-Woo Kim, **Hyun-Jung Kim**[†], Sangmo Cheon[†], and Tae-Hwan Kim[†], *Circular Dichroism of Emergent Chiral Stacking Orders in Quasi-One-Dimensional Charge Density Waves*, *Phys. Rev. Lett.* 128, 046401 (2022) [[†] **corresponding author**]
2. Shuyuan Liu, Chongze Wang, Liangliang Liu, Jin-Ho Choi, **Hyun-Jung Kim**, Yu Jia, Chul Hong Park, and Jun-Hyung Cho, *Ferromagnetic Weyl Fermions in Two-Dimensional Layered Electride Gd₂C*, *Phys. Rev. Lett.* 125, 187203 (2020).

3. Sun-Woo Kim*, Yoon-Gu Kang*, **Hyun-Jung Kim***, and Jun-Hyung Cho, *Comment on “Quasi-One-Dimensional Metal-Insulator Transitions in Compound Semiconductor Surfaces”*, *Phys. Rev. Lett.* 118, 239601 (2017). [***contributed equally**]
4. Sun-Woo Kim, Chen Liu, **Hyun-Jung Kim**, Jun-Ho Lee, Yongxin Yao, Kai-Ming Ho, and Jun-Hyung Cho, *Nature of the Insulating Ground State of 5d Post-Perovskite CaIrO_3* , *Phys. Rev. Lett.* 115, 096401 (2015).
5. Hui Zhang*, Fangfei Ming*, **Hyun-Jung Kim***, Hongbin Zhu, Qiang Zhang, Hanno H. Weitering, Xudong Xiao, Changgan Zeng, Jun-Hyung Cho, and Zhenyu Zhang, *Stabilization and Manipulation of Electronically Phase-Separated Ground States in Defective Indium Atom Wires on Silicon*, *Phys. Rev. Lett.* 113, 196802 (2014). [***contributed equally**]
→ *Selected as outstanding research achievement* (National Research Foundation, Grant No. 2014M2B2A9032247)
6. Jun-Ho Lee, **Hyun-Jung Kim**, and Jun-Hyung Cho, *Ferrimagnetic Slater Insulator Phase of the $\text{Sn/Ge}(111)$ Surface*, *Phys. Rev. Lett.* 111, 106403 (2013).
7. **Hyun-Jung Kim** and Jun-Hyung Cho, *Driving Force of Phase Transition in Indium Nanowires on $\text{Si}(111)$* , *Phys. Rev. Lett.* 110, 116801 (2013).

• **Physical Review B: Rapid Communication**

8. Sun-Woo Kim, **Hyun-Jung Kim**[†], Sangmo Cheon[†], and Tae-Hwan Kim[†], *Two-dimensional chiral stacking orders in quasi-one-dimensional charge density waves*, *Phys. Rev. B (Rap. Comm.)* 102, 121408 (2020). [**† corresponding author**]
9. **Hyun-Jung Kim**, Seoung-Hun Kang, Ikutaro Hamada, and Young-Woo Son, *Origins of the structural phase transitions in MoTe_2 and WTe_2* , *Phys. Rev. B (Rap. Comm.)* 95, 180101 (2017).
10. **Hyun-Jung Kim**, Chaokai Li, Ji Feng, Jun-Hyung Cho, and Zhenyu Zhang, *Competing magnetic orderings and tunable topological states in two-dimensional hexagonal organometallic lattices*, *Phys. Rev. B (Rap. Comm.)* 93, 041404 (2016).
→ *Selected as outstanding research achievement* (Korea Institute for Advanced Study)
11. **Hyun-Jung Kim**, Alexandre Tkatchenko, Jun-Hyung Cho, and Matthias Scheffler, *Benzene adsorbed on $\text{Si}(001)$: The role of electron correlation and finite temperature*, *Phys. Rev. B (Rap. Comm.)* 85, 041403 (2012).
12. **Hyun-Jung Kim**, Sangchul Oh, Ki-Seok Kim, Zhenyu Zhang, and Jun-Hyung Cho, *Length- and parity-dependent electronic states in one-dimensional carbon atomic chains on $\text{C}(111)$* , *Phys. Rev. B (Rap. Comm.)* 82, 041401 (2010). → *Selected in “Virtual Journal of Nanoscale Science & Technology”, Vol. 22, July 19 (2010).*

• **Physical Review B**

13. Seoung-Hun Kang, Sangjun Jeon, **Hyun-Jung Kim**, Wonhee Ko, Suyeon Cho, Se Hwang Kang, Hee-jun Yang, Hyo Won Kim, and Young-Woo Son, *Reshaped Weyl fermionic dispersions driven by Coulomb interactions in MoTe_2* , *Phys. Rev. B* 105, 045143 (2022).
14. Sang-Min Jeong*, Seho Yi*, **Hyun-Jung Kim***, Gustav Bihlmayer, and Jun-Hyung Cho, *Competing edge structures of Sb and Bi bilayers generated by trivial and nontrivial band topologies*, *Phys. Rev. B* 98, 075402 (2018). [***contributed equally**]

15. Sun-Woo Kim, Hyun Jung, **Hyun-Jung Kim**, Jin-Ho Choi, Su-Huai Wei, and Jun-Hyung Cho, *Microscopic mechanism of the tunable band gap in potassium-doped few-layer black phosphorus*, *Phys. Rev. B* 96, 075416 (2017).
16. Xiao-Yan Ren, **Hyun-Jung Kim**, Seho Yi, Yu Jia, and Jun-Hyung Cho, *Spin-orbit coupling effects on the stability of two competing structures in Pb/Si(111) and Pb/Ge(111)*, *Phys. Rev. B* 94, 075436 (2016).
17. **Hyun-Jung Kim** and Jun-Hyung Cho, *Giant spin-orbit-induced spin splitting in Bi zigzag chains on GaAs(110)*, *Phys. Rev. B* 92, 085303 (2015).
18. Sun-Woo Kim, **Hyun-Jung Kim**, Jin-Ho Choi, Ralph H. Scheicher, and Jun-Hyung Cho, *Contrasting interedge superexchange interactions of graphene nanoribbons embedded in h-BN and graphane*, *Phys. Rev. B* 92, 035443 (2015).
19. Sun-Woo Kim, **Hyun-Jung Kim**, Fangfei Ming, Yu Jia, Changgan Zeng, Jun-Hyung Cho, and Zhenyu Zhang, *Equivalence of electronic and mechanical stresses in structural phase stabilization: A case study of indium wires on Si(111)*, *Phys. Rev. B* 91, 174434 (2015).
20. **Hyun-Jung Kim** and Jun-Hyung Cho, *Fluorine-induced local magnetic moment in graphene: A hybrid DFT study*, *Phys. Rev. B* 87, 174435 (2013).
21. **Hyun-Jung Kim**, Sangchul Oh, and Jun-Hyung Cho, *Peierls instability and ferrimagnetic ordering of quasi-one-dimensional carbon chains generated in H-passivated graphene*, *Phys. Rev. B* 83, 235408 (2011).

• Physical Review Materials

22. Bharti Parashar, Lars Rathmann, **Hyun-Jung Kim**, Iulia Cojocariu, Aaron Bostwick, Chris Jozwiak, Eli Rotenberg, José Avila, Pavel Dudin, Vitaliy Feyer, Christoph Stampfer, Bernd Beschoten, Gustav Bihlmayer, Claus M. Schneider, and Lukasz Plucinski, *Photoemission study of twisted monolayers and bilayers of WSe₂ on graphite substrates*, *Phys. Rev. Mater.* 7, 044004 (2023).
23. Irene Aguilera, **Hyun-Jung Kim**, Christoph Friedrich, Gustav Bihlmayer, and Stefan Blügel, *Z₂ topology of bismuth*, *Phys. Rev. Mater.* 5, L091201 (2021).

American Chemical Society Journals (3)

24. Hyo Won Kim*, Seoung-Hun Kang*, **Hyun-Jung Kim***, Kisung Chae, Suyeon Cho, Wonhee Ko, Sangjun Jeon, Se Hwang Kang, Heejun Yang, Sung Wng Kim, Seongjun Park, Sungwoo Hwang, Young-Kyun Kwon, Young-Woo Son, *Symmetry Dictated Grain Boundary State in a Two-dimensional Topological Insulator*, *Nano Lett.* 20, 5837–5843 (2020) [*contributed equally]
25. Jun-Ho Lee, Yun-Ki Choi, **Hyun-Jung Kim**, Ralph H. Scheicher, and Jun-Hyung Cho, *Physisorption of DNA Nucleobases on h-BN and Graphene: vdW-Corrected DFT Calculations*, *J. Phys. Chem. C* 117, 13435–13441 (2013).
26. **Hyun-Jung Kim**, Sangchul Oh, Changgan Zeng, and Jun-Hyung Cho, *Peierls Instability and Spin Orderings of Ultranarrow Graphene Nanoribbons in Graphane*, *J. Phys. Chem. C* 116, 13795–13799 (2012).

Materials Science, Physical Chemistry, and Related Journals (7)

27. Yu Dai, Wenjiang Zhou, **Hyun-Jung Kim**, Qichen Song, Xin Qian, Te-Huan Liu, and Ronggui Yang, *Simultaneous enhancement in electrical conductivity and Seebeck coefficient by single- to double-valley transition in a Dirac-like band*, *npj Comput. Mater.* 8, 234 (2022).

28. Huije Ryu, Yangjin Lee, **Hyun-Jung Kim**, Seoung-Hun Kang, Yoongu Kang, Kangwon Kim, Jungcheol Kim, Blanka E. Janicek, Kenji Watanabe, Takashi Taniguchi, Pinshane Y. Huang, Hyeon-sik Cheong, In-Ho Jung, Kwanpyo Kim, Young-Woo Son, and Gwan-Hyoung Lee, *Anomalous Dimensionality-driven Phase Transition of MoTe₂ in van der Waals Heterostructure*, *Adv. Funct. Mater.* **31**, 2107376 (2021).
29. Seho Yi, Jin-Ho Choi, **Hyun-Jung Kim**, Chul Hong Park, and Jun-Hyung Cho, *Contrasting diffusion behaviors of O and F atoms on graphene and within bilayer graphene*, *Phys. Chem. Chem. Phys.* **19**, 9107–9112 (2017).
30. Xiao-Yan Ren, **Hyun-Jung Kim**, Chun-Yao Niu, Yu Jia, Jun-Hyung Cho, *Origin of Symmetric Dimer Images of Si(001) Observed by Low-Temperature Scanning Tunneling Microscopy*, *Sci. Rep.* **6**, 27868 (2016).
31. **Hyun-Jung Kim**, Jun-Ho Lee, and Jun-Hyung Cho, *Antiferromagnetic Slater Insulator Phase of Na₂IrO₃*, *Sci. Rep.* **4**, 5253 (2014).
→*Selected in Outstanding Paper Award* (Research Institute for Natural Sciences, Hanyang University, 2015)
32. Sun-Woo Kim, Jun-Ho Lee, **Hyun-Jung Kim**, and Jun-Hyung Cho, *Contribution of van der Waals interactions to the adsorption energy of C₂H₂, C₂H₄, and C₆H₆ on Si(100)*, *Chem. Phys. Lett.* **557**, 159–162 (2013).
33. **Hyun-Jung Kim**, Xingyong Wang, Jing Ma, and Jun-Hyung Cho, *A density-functional-theory study of biradicals from benzene to hexacene*, *Chem. Phys. Lett.* **516**, 141–145 (2011).

Peer-Reviewed Display Conference Proceedings — SID Display Week (1)

34. **Hyun-Jung Kim***, Junho Lee*, Yeol Kyo Choi, Taeyang Lee, Joong-Hwan Yang, Sung Moon Ko, Dae-Woong Jeong, Sehui Han, Jeongguk Min, Ji-Ho Baek, Seok-Woo Lee, Joon-Young Yang, Soo-Young Yoon, 85-3: *Machine Learning Strategy Towards Inverse Design of Blue TADF Emitter: Training Excited State Properties Based on Density Functional Theory Calculations*, *SID Symposium Digest of Technical Papers* **55**, 1183–1186 (2024). [***contributed equally**]

Selected Technical Talks and Presentations

1. *Introduction to Quantum Computing and Perspectives*
DX Tech Seminar, AI/Big Data Division, LG Display, Korea, February 2025.
2. *Machine Learning Strategy Towards Inverse Design of Blue TADF Emitter: Training Excited State Properties Based on Density Functional Theory Calculations*
SID Display Week 2024 Symposium, San Jose, USA, May 2024.
3. *GOETHE: Global Optimization Toolkit for Empirical Tight-Binding Model Hamiltonian Evaluation*
Laboratory Seminar, LG Display, Korea, September 2022.
4. *Tight-Binding Hamiltonian Parameterization Using Machine Learning*
Group Seminar, Universität Duisburg-Essen, Duisburg, Germany, June 2022.
5. *TBFIT: An Open-Source Software Package for Slater–Koster Tight-Binding Parameter Fitting*
Quantum Theory of Materials Seminar, Forschungszentrum Jülich, Germany, November 2018.

Appendix: Additional Presentation Record

Additional Conference Talks (23)

- **2019** — *An Open-Source Software Package for Slater–Koster Tight-Binding Parameterization and Calculation* — Korean Physical Society Spring Meeting, Korea.
- **2018** — *Weyl Fermion in the Surface States of Bi-Adsorbed Si(110) with a Nonsymmorphic Lattice* — Korean Physical Society Spring Meeting, Korea.
- **2018** — *Weyl Fermion on a Reconstructed Bi/Si(110) Surface with Non-Symmorphic Lattice Symmetry* — APS March Meeting, Los Angeles, USA.
- **2017** — *Origins of Structural Phase Transition in Bulk MoTe₂ and WTe₂* — Korean Physical Society Spring Meeting, Korea.
- **2017** — *A van der Waals Density Functional Study of Structural Phase Transition in Bulk MoTe₂ and WTe₂* — APS March Meeting, New Orleans, USA.
- **2016** — *A van der Waals Density Functional Study of Interlayer Interaction and Phase Stability in Layered MTe₂ (M=Mo, W)* — Korean Physical Society Fall Meeting, Korea.
- **2016** — *Spin–Valley Coupling and Topological Phase Transition in an Antiferromagnetic 2D Organometallic Thin Film* — International Conference on the Physics of Semiconductors, Beijing, China.
- **2016** — *Electric-Field-Induced Quantum Anomalous Hall Effect in a Two-Dimensional Antiferromagnetic Triphenyl-Lead Lattice* — APS March Meeting, Baltimore, USA.
- **2015** — *Electric-Field-Induced Topological Phase Transition in a 2D Organometallic Lattice* — International Conference on the Formation of Semiconductor Interfaces, Hiroshima, Japan.
- **2015** — *Giant Spin–Orbit-Induced Spin Splitting in Bi Zigzag Chains on GaAs(110)* — Korean Physical Society Fall Meeting, Korea.
- **2014** — *Structural Phase Transition in a Nanowire Controlled by Strain and Electric Field* — Korean Physical Society Spring Meeting, Korea.
- **2014** — *Structural and Electronic Properties of Indium Nanowires on Si(111)* — Workshop on Surface Nano-Science, Korea.
- **2013** — *Fluorine-Induced Local Magnetic Moment in Graphene: A Hybrid DFT Study* — Korean Vacuum Society Annual Conference, Korea.
- **2013** — *Driving Force of Phase Transition in Indium Nanowires on Si(111)* — International Conference on the Formation of Semiconductor Interfaces, Korea.
- **2013** — *Robustness of Topological Edge States Against Peierls Distortion and Reconstruction* — Korean Physical Society Spring Meeting, Korea.
- **2013** — *Driving Force of Phase Transition in Indium Nanowires on Si(111)* — Korean Physical Society Spring Meeting, Korea (Best Presentation Award).
- **2013** — *Robustness of Topological Edge States Against Peierls Distortion and Reconstruction* — Asian Workshop on Surface Nano-Science, Korea.
- **2012** — *Peierls Instability and Spin Orderings of Ultranarrow Graphene Nanoribbons in Graphane* — Korean Physical Society Spring Meeting, Korea.
- **2012** — *Peierls Instability and Spin Orderings in Graphane* — 42nd Annual Conference of the Korean Vacuum Society, Korea.
- **2012** — *Benzene Adsorbed on Si(001): The Role of Electron Correlation and Finite Temperature* — Asian Workshop on Surface Nano-Science, Korea.
- **2011** — *The Structure of Benzene on Si(001) at Zero and Finite Temperatures: Electronic Correlation and Vibrational Effects* — Korean Physical Society Fall Meeting, Korea.
- **2011** — *Electron Correlation Effect in the Adsorption of Benzene on Si(100) within van der Waals Density Functional and Random Phase Approximations* — Korean Physical Society Spring Meeting, Korea.
- **2010** — *Length- and Parity-Dependent Electronic States in One-Dimensional Carbon Atomic Chains on C(111)* — Korean Vacuum Society Annual Conference, Korea.

Additional Technical and Departmental Seminars (16)

- **2022** — *Circular Dichroism of Emergent Chirality in Quasi-One-Dimensional Charge Density Waves* — Quantum Theory of Materials Seminar, Forschungszentrum Jülich, Germany.
- **2022** — *Electronic Properties of Twisted WSe₂ Bilayers and WSe₂ Single Layer on Graphene* — Korea Institute for Advanced Study, Korea.
- **2022** — *Tight-Binding Approximations and Their Application to Bismuth and Transition-Metal Dichalcogenides* — Hanyang University, Korea.
- **2020** — *Two-Dimensional Chiral Stacking Orders in Quasi-One-Dimensional Charge Density Waves* — Quantum Theory of Materials Seminar, Forschungszentrum Jülich, Germany.
- **2019** — *Computational Approaches on Structural, Electronic, and Topological Properties in Low-Dimensional Systems* — Hanyang University, Korea.
- **2018** — *Weyl Fermion in the Surface States of Bi-Adsorbed Si(110)* — Quantum Theory of Materials Seminar, Forschungszentrum Jülich, Germany.
- **2018** — *Emergence of Massless Weyl Fermions in a Two-Dimensional System* — Computational Sciences Workshop, Korea Institute for Advanced Study, Korea.
- **2017** — *Structural, Electronic, and Topological Phase Transitions in Low-Dimensional Systems* — Dongguk University, Korea.

- **2015** — *Spin-Valley Coupling and Tunable Topological States in Two-Dimensional Antiferromagnetic Triphenyl-Lead Lattices* — UNIST, Korea.
- **2015** — *A First-Principles DFT Study of Tunable Topological States in Two-Dimensional Antiferromagnetic Triphenyl-Lead Lattices* — Computational Sciences Workshop, Korea Institute for Advanced Study, Korea.
- **2015** — *Giant Rashba-Type Spin Splitting in One-Dimensional Bi Chains on GaAs(110)* — Korea Institute for Advanced Study, Korea.
- **2015** — *Structural/Electronic Stability in a Valley-Polarized AFM State in a 2D Metal-Organic Framework* — ICQD Brainstorming Session, University of Science and Technology of China, China.
- **2013** — *Electronic Structure in One-Dimensional Systems: Graphene Nanoribbon in Graphene, In/Si Nanoarray, and Bi(111) Nanoribbon* — ICQD Brainstorming Session, University of Science and Technology of China, China.
- **2011** — *Benzene Adsorbed on Si(001): The Role of Electron Correlation and Finite Temperature* — Hanyang University, Korea.
- **2011** — *Competing Charge and Spin Orderings of One-Dimensional Carbon Chains* — Forschungszentrum Jülich, Germany.
- **2011** — *A First-Principles Study of Benzene/Si(100): Role of van der Waals Interactions* — Fritz Haber Institute, Germany.

Poster Presentations (21)

- **2020** — *The Role of Strain and Interlayer Interaction in Layered Higher-Order Topological Insulators: Multi-Scale Numerical Simulations* — Alexander von Humboldt Foundation Virtual Network Meeting.
- **2020** — *TBFIT: An Open-Source Software Package for Slater-Koster Tight-Binding Parameter Fitting* — Workshop on Computational Physics and Materials Science, Donostia-San Sebastián, Spain.
- **2019** — *Six-Nearest-Neighbor Parameterized Slater-Koster Tight-Binding Model for Bismuth* — Asian Workshop on First-Principles Electronic Structure Calculations, Osaka, Japan.
- **2019** — *Six-Nearest-Neighbor Parameterized Slater-Koster Tight-Binding Model for Bi* — Recent Progress in Graphene and 2D Materials, Matsue, Japan.
- **2019** — *TBFIT: An Open-Source Software Package for Slater-Koster Tight-Binding Parameter Fitting* — Automated Interactive Infrastructure and Database for Computational Science, EPFL, Switzerland.
- **2019** — *Nodal Fermion in the Surface States of a Bi-Adsorbed Semiconductor Surface* — Korean Vacuum Society Winter Conference, Korea.
- **2018** — *Adsorption Structure of a Bi-Adatom-Induced Si(110)-(3×2) Surface* — Korean Vacuum Society Winter Conference, Korea.
- **2017** — *Intertwined Structural and Electronic Phase Transition in Layered MoTe₂ and WTe₂* — Asian Workshop on First-Principles Electronic Structure Calculations, Nanjing, China.
- **2017** — *Contrasting Structural and Electrical Phase Transition Behaviors of Layered MoTe₂ and WTe₂* — Recent Progress in Graphene and Two-Dimensional Materials Research, Singapore.
- **2016** — *Interlayer Interaction and Phase Stability in Bulk MoTe₂ and WTe₂: A van der Waals Density Functional Study* — Asian Workshop on First-Principles Electronic Structure Calculations, Hsinchu, Taiwan.
- **2016** — *Electric-Field-Induced Quantum Anomalous Hall Effect in a Spin-Valley-Coupled 2D Triphenyl-Lead Monolayer* — USTC-FHI Workshop on Advanced Electronic Structure Methods, Hefei, China.
- **2015** — *Magnetic Ordering and Topological Phase Transition in a 2D Organometallic Thin Film: Pb₂[C₃H₅]₃ Lattice* — Asian Workshop on First-Principles Electronic Structure Calculations, Tokyo, Japan.
- **2015** — *Tunable Topological States in a Hexagonal Triphenyl-Lead Lattice* — International Conference on Recent Progress in Graphene and Two-Dimensional Materials Research, Victoria, Australia.
- **2015** — *Electronic Properties in a 2D Hexagonal Triphenyl-Lead Lattice and Topological Phase Transition via Applied Electric Field* — Korea Supercomputing Conference, Korea.
- **2014** — *Strain- and Electric-Field-Controlled Phase Transition in Indium Atom Wires on Silicon* — International Conference and Exhibition for Nanopia, Korea.
- **2014** — *Strain and Electric-Field Effects on Phase Transition in an Indium Atomic Wire on Silicon* — IPAM Graduate Summer School, Los Angeles, USA.
- **2014** — *Metal-Insulator Phase Transition in Indium Nanowires on Si(111)* — 6th HOPE Meeting, Tokyo, Japan.
- **2014** — *Effects of Self-Interaction Correction and Spin-Orbit Coupling on the Insulating Phase of Na₂IrO₃* — Quantum Materials Symposium, Korea.
- **2013** — *Role of Strain and Electric Field in the Phase Transition of an Indium Nanowire on Si(111)* — Korean Physical Society Fall Meeting, Korea.
- **2012** — *Influence of Geometry on the Edge States of Bi Nanoribbons* — International Conference on Magnetism, Korea.
- **2010** — *Influence of Spin Delocalization on the Singlet-Triplet Gap of Diradicals Generated in Polyacenes: Density-Functional Theory Calculations* — Korean Physical Society Fall Meeting, Korea.